

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

Source: `results/gpt-5.6-sol-xhigh/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

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Approaches tried

- **Angle-coordinate model (successful):** If the current angles are (A, B, C) and a cut from the A -vertex splits A into x and $A - x$, the children have angles $(B, x, 180^\circ - B - x)$ and

$(C, A - x, B + x)$. Thus the two new angles at the cut point are supplementary.

- **Finite-horizon backward sets (partially useful):** Starting from triangles already containing θ , an attempted full description of successive winning sets became unwieldy. Its useful consequence was the recursive observation that every triangle containing $m\theta$, for a positive integer m , is winning.
- **Complementary cut-point angles (successful):** When $180^\circ = n\theta$, a cut can be chosen so that the two children have cut-point angles $k\theta$ and $(n - k)\theta$. An elementary integer-between-partial-sums lemma guarantees such a legal cut from every triangle not already containing an integral multiple of θ .
- **Safe-triangle invariant (successful):** Call a triangle safe if none of its angles is a positive integral multiple of θ . If $180^\circ/\theta$ is not an integer, every cut of a safe triangle has a safe child. Starting from an equilateral safe triangle, Shan-Yu can therefore avoid defeat forever.
- **Sanity checks:** `code/sanity_checks.py` checked the safe-child invariant on 130242 rational-grid cut configurations and the integer interval lemma on 18744 rational-grid triples. These computations are not used as proof.

Current best

The complete characterization is

$$\theta = \frac{180^\circ}{n} \quad \text{for an integer } n \geq 2.$$

For these and only these values Mulan has a finite winning strategy. In fact, the proof gives a bound depending only on n for the number of cuts.

Full proof

We prove both directions. All angles below are measured in degrees.

1. The effect of a cut

Let a triangle ABC have angles A, B, C . Suppose that P is an interior point of BC , and put

$$x = \angle BAP.$$

Then $0 < x < A$. The angle triples of the two resulting triangles ABP and ACP are, respectively,

$$(B, x, 180^\circ - B - x) \quad (1)$$

and

$$(C, A - x, B + x). \quad (2)$$

Indeed, the first two entries displayed in each triple are the inherited angles at the original vertices, and the third follows from the angle sum of a triangle, using $A + B + C = 180^\circ$.

Conversely, every x with $0 < x < A$ can be realized by a legal cut: the ray from A inside $\angle BAC$ which makes angle x with AB meets the opposite side BC at an interior point. Thus, in using (1)–(2), it is enough to verify $0 < x < A$.

2. Triangles containing integral multiples of θ

We first establish a forcing observation.

Lemma 1. If the current triangle has an angle $m\theta$, where m is a positive integer, then Mulan can force a win after at most $m - 1$ further cuts.

Proof. We use induction on m . For $m = 1$, the stopping condition already holds, so no cut is needed.

Let $m \geq 2$. From the vertex with angle $m\theta$, Mulan makes the cut which divides that angle into

$$\theta + (m - 1)\theta.$$

Both parts are positive, so this is a legal cut by the observation above. One child has an angle θ , while the other has an angle $(m - 1)\theta$. If Shan-Yu retains the first, Mulan wins at the next stopping check. If he retains the second, the induction hypothesis gives a win after at most $m - 2$ additional cuts. In either case at most $m - 1$ cuts are required. $\square$

3. Sufficiency

Suppose

$$180^\circ = n\theta \quad (3)$$

for an integer $n \geq 2$. We show that Mulan can win from every initial triangle.

If the current triangle has an angle equal to a positive integral multiple of θ , Lemma 1 applies. It remains to handle a triangle none of whose angles is an integral multiple of θ .

Divide its three angles by θ , obtaining positive, nonintegral real numbers a, b, c such that

$$a + b + c = n. \quad (4)$$

We need the following elementary claim.

Claim. After possibly relabeling a, b, c , there is an integer k such that

$$b < k < a + b. \quad (5)$$

Moreover, $1 \leq k \leq n - 1$.

Proof of the claim. First suppose at least one of a, b, c is greater than 1. Name such a number a , name either of the remaining numbers b , and take $k = \lceil b \rceil$. Because b is not an integer,

$$0 < k - b < 1 < a,$$

which proves (5).

Otherwise all three numbers are less than 1: they are not equal to 1 because they are nonintegral. Hence (4) gives $n < 3$. Since $n \geq 2$ is an integer, $n = 2$. Name the numbers in any order. We have $b < 1$, while

$$a + b = 2 - c > 1.$$

Thus $k = 1$ satisfies (5).

Finally, in either case, (5) and positivity give $k > b > 0$, and

$$k < a + b = n - c < n.$$

Since k is an integer, $1 \leq k \leq n - 1$. This proves the claim. $\square$

Label the original triangle so that its angles are

$$A = a\theta, \quad B = b\theta, \quad C = c\theta,$$

where (5) holds. Mulan cuts from the vertex with angle A , choosing

$$x = (k - b)\theta.$$

The inequalities (5) give

$$0 < x < a\theta = A,$$

so this is a legal cut. By (1)–(2), one child has at the cut point the angle

$$B + x = b\theta + (k - b)\theta = k\theta, \quad (6)$$

while the other has at the cut point the angle

$$180^\circ - B - x = n\theta - k\theta = (n - k)\theta. \quad (7)$$

Both k and $n - k$ are positive integers. Whichever child Shan-Yu retains, it therefore contains a positive integral multiple of θ , so Lemma 1 gives Mulan a win after finitely many further cuts. This proves sufficiency.

4. Necessity

Now suppose that

$$\frac{180^\circ}{\theta} \notin \mathbb{Z}. \quad (8)$$

Call a triangle **safe** if none of its angles is a positive integral multiple of θ . We show that after every cut of a safe triangle, at least one child is safe.

Let the safe parent have angles A, B, C , and consider a cut from the vertex with angle A . With the notation of (1)–(2), its children have angle triples

$$(B, x, 180^\circ - B - x), \quad (C, A - x, B + x). \quad (9)$$

Assume for contradiction that both children are not safe. The inherited angles B and C are not integral multiples of θ , because the parent is safe. Hence there are positive integers p, q such that one equality from each of the following rows holds:

$$x = p\theta \quad \text{or} \quad 180^\circ - B - x = p\theta, \quad (10)$$

$$A - x = q\theta \quad \text{or} \quad B + x = q\theta. \quad (11)$$

There are four possible pairings:

1. If $x = p\theta$ and $A - x = q\theta$, then

$$A = (p + q)\theta,$$

contradicting safety of the parent.

2. If $x = p\theta$ and $B + x = q\theta$, then

$$B = (q - p)\theta.$$

Since $B > 0$, $q - p$ is a positive integer, again contradicting safety.

3. If $180^\circ - B - x = p\theta$ and $A - x = q\theta$, then, using $A + B + C = 180^\circ$,

$$C = (180^\circ - B - x) - (A - x) = (p - q)\theta.$$

Since $C > 0$, this contradicts safety.

4. If $180^\circ - B - x = p\theta$ and $B + x = q\theta$, then adding gives

$$180^\circ = (p + q)\theta,$$

contradicting (8).

All possibilities are impossible. Therefore at least one child of every cut of a safe triangle is safe.

Finally, Shan-Yu can choose an equilateral initial triangle. It is safe: if its angle 60° were $m\theta$ for some positive integer m , then

$$\frac{180^\circ}{\theta} = 3m$$

would be an integer, contrary to (8). After every cut, Shan-Yu retains a safe child, which exists by the preceding argument. Every retained triangle is nondegenerate because the cut point is not a vertex. Since a safe triangle has no angle θ , the stopping condition never occurs. Thus Mulan cannot guarantee a finite victory when (8) holds.

Combining sufficiency and necessity, and observing that $0^\circ < \theta < 180^\circ$ makes an integral value $180^\circ/\theta$ at least 2, yields exactly

$$\boxed{\theta = 180^\circ/n \quad (n = 2, 3, 4, \dots)}.$$